

Twitter Sentiment Analysis using NLP

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Abstract

This project explores Natural Language Processing (NLP) methods to detect hate speech within Twitter posts. We present a complete workflow: data acquisition, pre-processing, feature engineering, multiple machine learning models, and evaluation, with a focus on class imbalance as is common in real-world NLP applications. The report emphasizes both technical steps and their motivation within sentiment analysis.

1 Introduction

Natural Language Processing (NLP) has revolutionized textual data analysis. Twitter, as a global platform, sees millions of user posts daily, and is frequently an origin for both opinion mining and the spread of harmful content. Hate speech on social media poses not only technical but significant societal challenges. Manual moderation is infeasible at scale, highlighting the need for automated tools capable of detecting offensive content in real time.

Recent advances in NLP, including tokenization, vectorization, and supervised learning, allow for robust sentiment analysis and can be leveraged for tasks as nuanced as hate speech detection.

2 Objective and Scope

The objective is to build an NLP pipeline to classify tweets as containing hate speech (racist/sexist) or not. Specific goals:

- Preprocess raw Twitter data for NLP modeling.
- Engineer features that capture the sentiment and context of each tweet.
- Train and compare multiple classification models (Logistic Regression, SVC, Random Forest, Decision Tree).

- Address class imbalance—a frequent challenge in hate speech datasets—emphasizing robust evaluation.

3 Dataset Description

We use a publicly available dataset of **31,962 training tweets** and **17,197 test tweets**, each sample labeled as hate speech (1) or not (0). Labels are highly imbalanced, with the majority of tweets being non-hate. **Key columns:**

- **tweet:** The text content.
- **label:** 0 (not hate speech), 1 (hate speech).

Missing value checks and tweet length analysis were performed, confirming the data’s integrity.

4 Data Preprocessing

Careful preprocessing improves NLP pipeline effectiveness:

1. **Remove Twitter Handles:** Strips user mentions (e.g., @user).
2. **Remove Punctuation/Numbers/Special Characters:** Retains only meaningful words.
3. **Tokenization:** Splits text into tokens/words.
4. **Remove Stopwords:** Eliminates filler words (e.g., 'the', 'is').
5. **Lemmatization/Stemming:** Converts words to their base forms.

5 Exploratory Data Analysis (EDA)

EDA reveals:

- **Label Distribution:** Dataset contains a much larger number of non-hate tweets.
- **Tweet Length:** Hate speech tweets tend to be longer.
- **Frequent Words:** Visualized using word clouds and bar charts, aiding feature selection and class insight.

6 Feature Engineering

Effective feature engineering is the backbone of text classification:

- **Bag-of-Words (BoW):** Vectorizes tweets by word occurrence.
- **TF-IDF:** Assigns higher weights to rare, informative words.

We found TF-IDF to consistently improve classifier performance and generalization.

7 Model Development

We evaluated several algorithms suitable for high-dimensional NLP features:

- Logistic Regression
- Support Vector Classifier (SVC)
- Decision Tree
- Random Forest

The dataset was split into training and validation sets. All models were evaluated using F1-score, a critical metric for imbalanced data.

8 Results and Analysis

Model	Train Accuracy	Val. Accuracy	F1 Score
Logistic Regression	0.9991	0.9312	0.536
SVC	0.9781	0.9523	0.4993
Decision Tree	0.9916	0.9312	0.5359
Random Forest	0.9991	0.9526	0.65

Table 1: Model Evaluation Results

Random Forest achieved the highest F1-score, showing best performance for hate speech detection. Analysis of confusion matrices highlights that most classification errors are false negatives (hate speech missed), showing the subtlety and challenge of online hate detection.

9 Conclusion

This NLP project demonstrates the feasibility of automated hate speech detection in real-world Twitter data. By carefully addressing text preprocessing, feature extraction, and the issue of class imbalance with proper metrics, we produced a robust classifier fit for real-world deployment.

10 Future Work

- Apply advanced NLP embeddings (e.g., BERT, Word2Vec).
- Investigate neural network-based models.
- Deploy the classifier in real-time environments.
- Explore multilingual and multimodal hate speech datasets.

11 References

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